

INSY 5337 – Spring 2025

Assignment 4

Assignment Title: Building a Data Pipeline for Retail Sales Analysis Using Databricks Free Edition

Objective: This assignment provides hands-on experience building an end-to-end data pipeline using Databricks. Students will ingest, clean, transform, store, and analyze retail sales data while applying foundational data engineering concepts such as Delta Lake, data quality enforcement, partitioning, and query optimization.

Deliverables:

1. Python/SQL Notebook implementing all pipeline steps with comments.
2. Short report (max 500 words) summarizing:
 - Pipeline design
 - Challenges encountered
 - Key insights from the data

Assignment Tasks:

0. Environment Setup

- Launch Databricks Free Edition.
- Create or attach a serverless cluster.
- Create a new notebook for the assignment.

1. Data Ingestion

Dataset:

- Format: CSV
- Columns:
TransactionID, ProductID, StoreID, QuantitySold, SalesAmount, TransactionDate

Tasks:

1. Download the provided retail sales dataset.
2. Upload the CSV file into Databricks Volumes (under the default database).
3. Use PySpark or SQL to read the file and write it as a Bronze Delta table:
 - Table name: Retail_Sales_Data

- Apply an explicit schema during ingestion. Note that this data can have corrupted entry so make sure you select a datatype that allows you can write all data in Bronze table.

2. Data Transformation

Apply the following transformations to produce a clean and analytic-ready dataset. As you write silver tables make sure you use a data type in your silver table schema that match the actual data type on the transformed table.

(a) Clean Null and Corrupted Rows

- Remove records containing missing or invalid values in required fields (TransactionID, ProductID, StoreID, QuantitySold, SalesAmount, TransactionDate).

(b) Deduplicate Transactions

- Identify and remove duplicate TransactionID entries.

(c) Validate Business Rules

Apply the following checks and filter out invalid records:

- QuantitySold > 0
- SalesAmount > 0
- TransactionDate must be a valid and parseable timestamp

(d) Required Analytical Transformations

Perform:

1. Standardize TransactionDate into yyyy-MM-dd.
2. Compute daily total sales per store.
3. Identify the most-sold product per store (based on total QuantitySold).
4. Identify and count stores with average daily sales > \$300.

(e) Store the transformed data

Write the transformed outputs into Silver Delta tables:

- Daily_Sales_Summary
- Top_Products_By_Store

3. Data Storage & Optimization

(a) Partitioning

Partition the transformed Delta tables by TransactionDate or StoreID based on whichever yields better query performance.

(b) Z-Ordering for Optimization

Use Delta Lake optimization features: OPTIMIZE and ZORDER; Explain your partitioning and ZORDER choices in the report.

4. Data Analysis

Using SQL in your notebook:

1. Retrieve the total sales for the top-performing store.
2. List the top 5 products across all stores by total sales.
3. Identify the highest and lowest sales dates for each store.

5. Data Visualization

Using Databricks built-in visualization tools, create:

1. Bar Chart: Total daily sales by store
2. Line Graph: Sales trends over one week for store number 10.

Evaluation Criteria/Grading:

Assignment Deliverables	Component	Points	Submission
1. Notebook	1. Data Ingestion	5	Notebook
	2. Data Transformation	25	
	3. Data Storage	15	
	4. Data Analysis - SQL Queries	20	
	5. Visualization	15	
2. Report	Explanation / Report	20	Report .pdf document

Submission Instructions:

1. Submit the completed Databricks notebook
2. Submit the report as a PDF file.

Deadline:

All submissions are due by December 6th, at 11:59 PM.